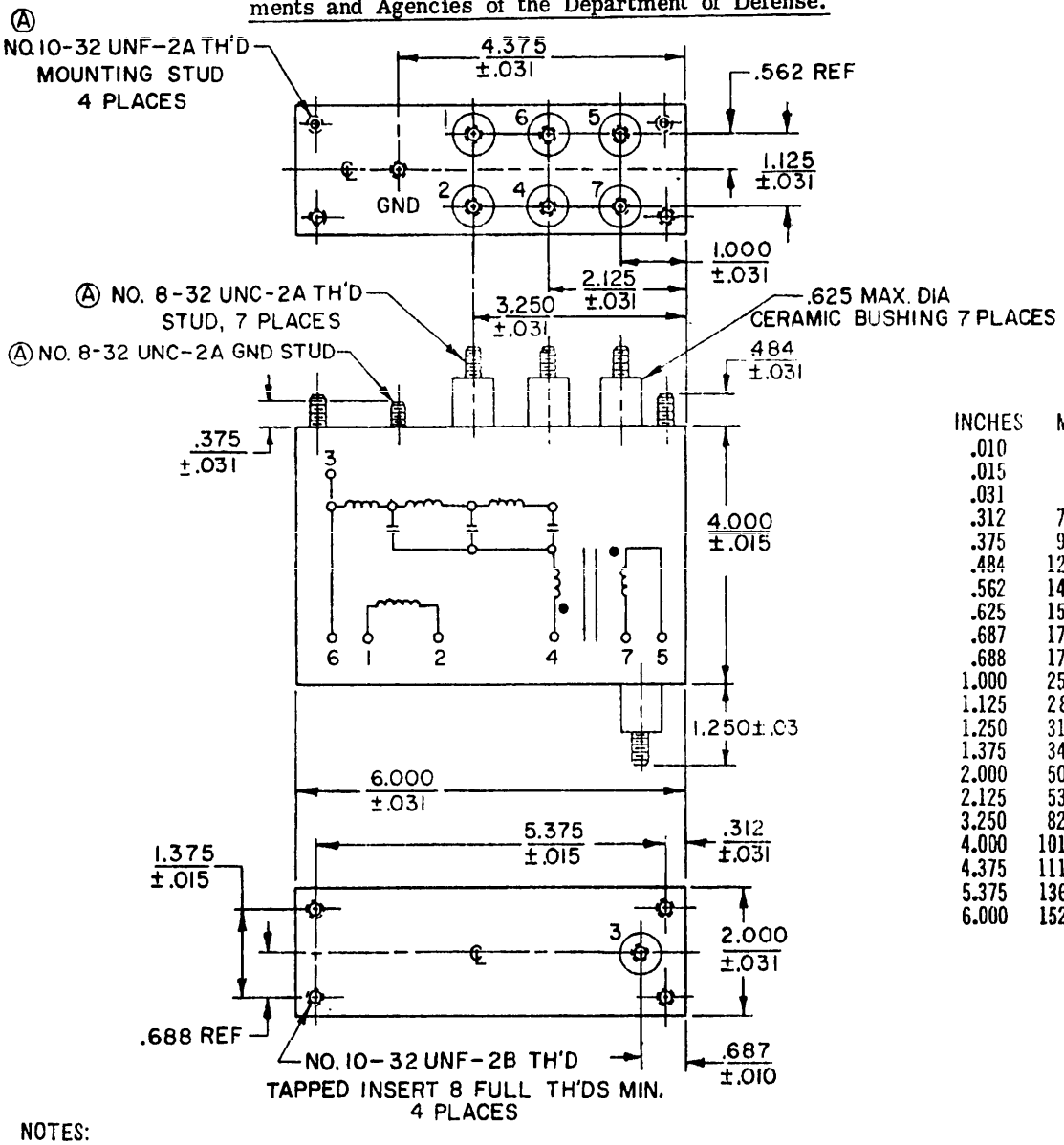
MILITARY SPECIFICATION SHEET

NETWORK, PULSE FORMING, RADAR, TYPE PN3E202C262Q500

The complete requirements for procuring the pulse forming network described herein shall consist of this document and the latest issue of Specification MIL-N-23182.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



INCHES	MM
.010	.25
.015	.38
.031	.79
.312	7.92
.375	9.53
.484	12.29
.562	14.27
.625	15.88
.687	17.45
.688	17.48
1.000	25.40
1.125	28.58
1.250	31.75
1.375	34.93
2.000	50.80
2.125	53.98
3.250	82.55
4.000	101.60
4.375	111.13
5.375	136.53
6.000	152.40

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm

Case and mounting dimensions, and circuit diagram.

Ⓐ denotes changes

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REQUIREMENTS:

Dimensions and configuration: See figure.

Case: Metal.

Circuit diagram: See figure.
Note - This item consists of a pulse forming network (PFN), charging reactor (CR), and pulse transformer (PT) in one container.

Weight: 6 pounds.

Maximum operating temperature: +85° C.

Pulse repetition frequency:
Maximum: 2,600 pulses per second.
Average: 1,710 pulses per second.
Minimum: 250 pulses per second.

Total capacitance (PFN):
0.0198 ± 0.0017 microfarad (uf).

Load impedance:
Pulse forming network: 50 ohms.
Pulse transformer:
Test: 50 ohms (match).
Maximum: 5948/1754 grid, shunted by 250 ohms.
Minimum: 10 ohms.

Supply voltage (to PFN): 560 vdc.

Charging voltage (PFN): 3,000 vdc.

Operating voltage:
(CR): 3,000 vdc, maximum.
(PT): 1,500 vdc, maximum.

Terminals: Screw.

Charging reactor:
Inductance: 0.8 henry (h) ±0.1 h.
DC resistance: 100 ohms, maximum.
Terminal No. 2: 20 picofarads (pf) capacity to ground; normal operating mode connected to load.
Terminal No. 1: Connected to 560 vdc.
Duty cycle: Continuous pulse operation.
Temperature rise: 30° C maximum.

Pulse transformer:
Turns ratio: 1:1.
Duty cycle: Continuous pulse operation.
Capacitance in shunt with primary and secondary windings of transformer due to case: Not to exceed 50 pf.

Terminal strength: Method 211, test condition E, of MIL-STD-202. The following detail shall apply:
Torque: 11 pound-inches.

Sealing: As specified in MIL-N-23182. Units shall be immersed for 2 to 3 minutes in a bath of water or any other liquid of no greater viscosity maintained at a temperature of at least +85° C. The temperature of the units shall not exceed +40° C at the time of immersion. During the immersion, units shall show no flow of air or gas bubbles (other than an occasional entrapped air or gas bubble), or leakage of compound.

Corona discharge: As specified in MIL-N-23182, at atmospheric pressures of from 29.92 inches of mercury (sea level) to 20.58 inches of mercury (10,000 feet) with no indication of corona.

Dielectric withstanding voltage: As specified in MIL-N-23182. The following details and exceptions shall apply:
Magnitude of test potential and points of application:
Pulse forming network: As specified in MIL-N-23182.
Charging reactor:
7,000 vdc between terminals 1 and 2.
7,000 vdc between terminals (1 and 2 tied together) and the container.
The pulse forming network and charging reactor shall withstand the specified potentials without breakdown, puncturing of insulation, or arcing.

Barometric pressure (reduced): Method 105 of MIL-STD-202. The following details and exceptions shall apply:
Atmospheric pressure: 20.58 inches of mercury (10,000 feet).
Application of test voltage:
Charging reactor: 125 percent of maximum operating voltage for at least 1 minute between each terminal and every other terminal in turn, and between the container and each terminal not connected to the container.

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Pulse transformer:
Same as for the charging reactor, except that maximum operating voltage shall be 1,500 vdc.
Pulse forming network:
Same as for the charging reactor, except that the test voltage shall be 125 percent of the rated charging voltage, and shall not be applied between terminals 3 and 6.
After the test, units shall be visually examined for evidence of short circuiting, arcing, breakdown of insulation and other visible damage.

Insulation resistance (PFN):
Method 302, test condition B, of MIL-STD-202.
The following details and exceptions shall apply:
Electrification time: 1 minute, maximum.
Terminal to terminal (terminal 3 to terminal 4).
At 25° C: 500 megohm-microfarads, minimum.
At +85° $^{+3°}_{-0°}$ C; 15 megohm-microfarads, minimum.
Terminals to container (where container is not a terminal), at 25° C: Shall exceed 3,000 megohms.

Dissipation factor at 25° C((PFN):
As specified in MIL-N-23182. The following details and exceptions shall apply:
Frequency of measurement (or reference): 1,000 Hz.
Test potential: 500 volts root mean square (rms), maximum.
Dissipation factor: Shall not exceed 1 percent.

Capacitance (PFN) (terminal to terminal):
Method 305 of MIL-STD-202. The following details and exceptions shall apply:
Test frequency: 1,000 Hz ± 100 Hz.
Test potential: 500 volts rms, maximum.
Limit of accuracy of measurement: ±2 percent.
Total measured capacitance: Within the limits of 0.0181 to 0.0215 uf.

Waveshape electrical characteristics:
Measured as specified in MIL-N-23182 (with exceptions noted below) and shall be:
Pulse forming network:
Pulse duration (at 70 percent amplitude points):
2.0 microseconds (μsec)., $^{+0.2}_{-0.0}$ μsec.
Percent ripple: 5 percent, maximum.
Pulse rise time: 0.20 μsec., maximum.
Pulse decay time: 0.40 μsec., maximum.
Characteristic impedance: 50 ohms.
Duty cycle (at 1,710 pulses per second): Between 0.00342 and 0.00376.
Pulse transformer:
Pulse duration: Same as for pulse forming network.
Pulse rise time: Same as for pulse forming network.
Pulse decay time: Same as for pulse forming network.

Early life survival:
As specified in MIL-N-23182. The following details and exceptions shall apply:
Prior to the test, units shall be subjected to requirements, at standard test conditions, for corona discharge and capacitance.
Terminating resistance (PFN):
Not to exceed 37.5 ohms.
Test temperature: Not less than +85° C.
Pulse repetition frequency: 428 pulses per second.
At the conclusion of the 24-hour exposure, units shall be subjected to and shall comply with requirements, at standard test conditions, for corona discharge. In addition, capacitance shall be measured at 25° C. Capacitance change, from the initial measurement, shall not exceed ±2 percent.

Thermal shock:
Method 107, test condition A, of MIL-STD-202.
The following details and exceptions shall apply:
Exposure time:
At -55° $^{+0°}_{-3°}$ C: 2 hours.
At +85° $^{+3°}_{-0°}$ C: 2 hours.
After completion of the final cycle, units shall be examined for evidence of leakage or other visible damage. In addition, units shall be subjected to and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insulation resistance, capacitance, and waveshape electrical characteristics.

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Moisture resistance:
Method 106, figure 106-1, of MIL-STD-202. The following details and exceptions shall apply:
Method of mounting: On racks by normal mounting means.
Conditioning: Units shall undergo a 24-hour initial drying period prior to first cycle.
Polarization voltage: 100 vdc, during steps 1 to 6 inclusive, between all terminals not connected directly to each other.
Upon completion of step 6 of the final cycle, units shall be removed from the humidity chamber and shall be examined for evidence of leakage, or other visible damage. Units shall then be subjected to and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insulation resistance, capacitance, and waveshape electrical characteristics.

Vibration:
Method 201 of MIL-STD-202. The following details and exceptions shall apply:
Method of mounting: By normal means.
Planes of vibration: As specified in MIL-N-23182 for units weighing 10 pounds or less.
Measurements during vibration:
During the last one-half hour of vibration, a suitable measurement shall be made to detect intermittent contacts, or open- or short-circuiting.
After the test, units shall be examined for evidence of leakage, or mechanical damage. Units shall then be subjected to and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insulation resistance, and capacitance.

Shock (medium impact):
Method 205, test condition B, of MIL-STD-202. The following details and exceptions shall apply:
Method of mounting: By normal means.
Number and direction of applied shocks: 30 impact shocks, 10 in each of three mutually perpendicular planes.
After the test, units shall be examined for evidence of physical or mechanical damage. In addition, units shall be subjected to and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insulation resistance, and capacitance.

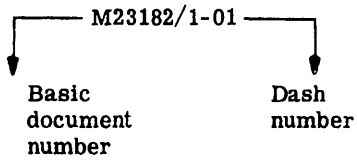
Low temperature:
As specified in MIL-N-23182. Units shall be subjected to a temperature of $-55\ ^{+0^\circ}_{-3^\circ}$ C for 48 hours with rated charging voltage applied, with no evidence of breakdown.

Capacitance change at reduced temperature:
As specified in MIL-N-23182. The following details and exceptions shall apply:
At the conclusion of the low temperature test, and while units are still subjected to the -55° C temperature, the capacitance of the pulse forming network shall be measured. The measurement shall be made at, or referred to, a frequency of 1,000 Hz. The test potential shall be 500 volts rms or less. Capacitance change shall not exceed -5 percent.

Salt spray (corrosion):
Method 101, test condition B, of MIL-STD-202. The following details and exceptions shall apply:
Salt solution concentration: 5 percent.
After the test, units shall be examined for evidence of corrosion and obliteration of marking.

Life:
As specified in MIL-N-23182 for networks with special life requirements. The following details and exceptions shall apply:
Exposure time: 1,000 hours.
Exposure temperature: $+85\ ^{+3^\circ}_{-0^\circ}$ C.
Test potential: 3,000 vdc.
Input power: 616 volt-amperes.
Temperature rise, as specified in MIL-N-23182, shall not exceed 30° C.
After the test, units shall be subjected to, and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insulation resistance, and waveshape electrical characteristics. In addition, units shall be examined for evidence of physical or electrical damage.

Part number:
Consists of the basic number of this specification sheet and a dash number.



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Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - EC
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - MI, SM
Navy - MC, AS, OS
Air Force - 19

Preparing activity:
Navy - EC

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